Chest X-Ray Pneumonia Detection Using Convolutional Neural Networks and Transfer Learning

Elif Sila Okutucu

MSc Data Science | University of Europe for Applied Sciences

Custom CNN | EfficientNetB0 | ResNet50 | Grad-CAM | Class Weighting

📊 Kaggle Chest X-Ray (Pneumonia) — 5,856 Images

Problem Statement & Research Questions

The Problem

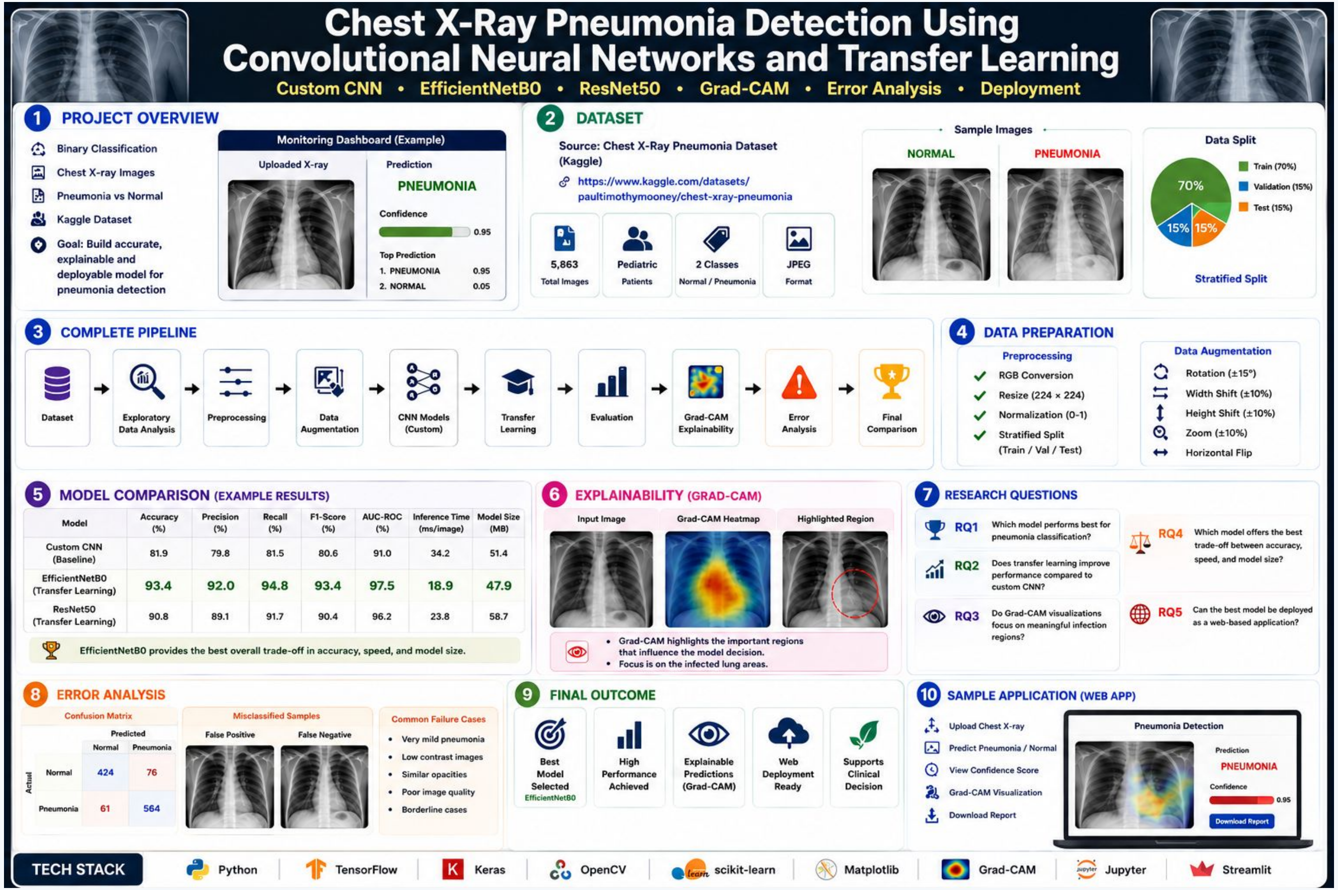
- Pneumonia kills millions annually — especially children under 5 and elderly
- Manual X-ray reading is time-consuming and inconsistent between clinicians
- Shortage of radiologists in low-income countries delays life-saving diagnosis
- Class imbalance (73% Pneumonia / 27% Normal) is rarely addressed in literature
- Few studies combine class weighting + Grad-CAM + error analysis together

7 Research Questions

- **RQ1** How effectively can a custom CNN classify pneumonia from chest X-ray images?
- **RQ2** To what extent do transfer learning models outperform a custom CNN across accuracy, recall, F1-score, and AUC-ROC?
- **RQ3** Can class weighting mitigate the impact of class imbalance on false negative predictions?
- **RQ4** How do evaluation metrics affect pneumonia classification assessment?
- **RQ5** What insights can Grad-CAM provide about model decision-making?
- **RQ6** Does data augmentation improve model generalization?
- **RQ7** What patterns appear in misclassified chest X-ray images?

Project Workflow

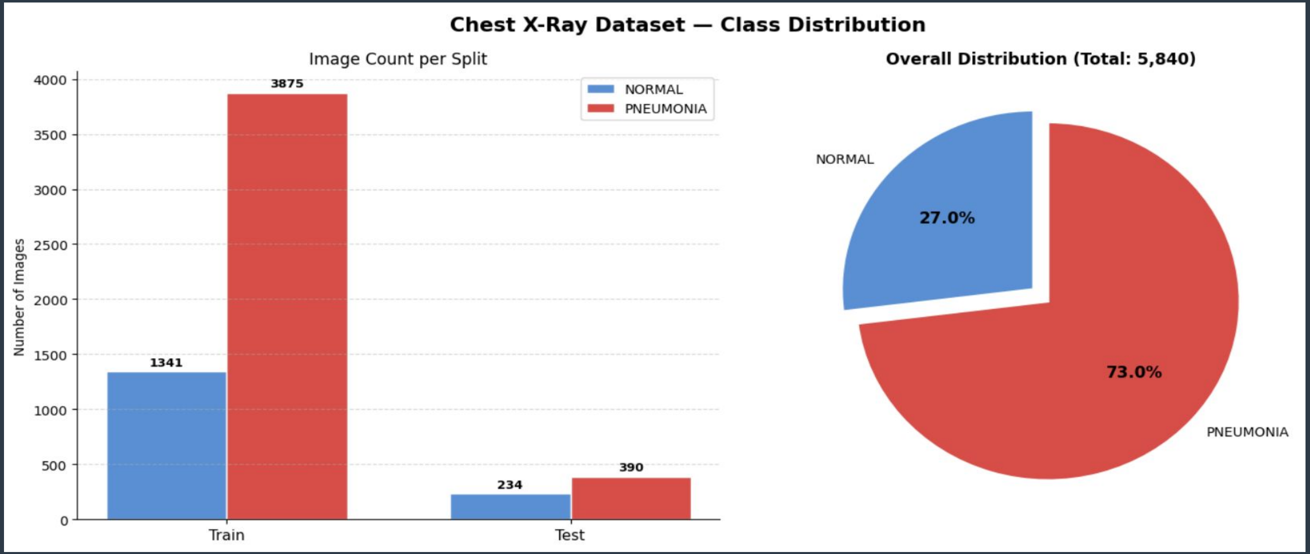
Here, the model comparison part gives indicative values, final results on held-out test set (n=624) are presented in the following results part. Additionally, corrected data split values given on the next page.



Dataset & Class Distribution

Kaggle Chest X-Ray (Pneumonia) — Paul Mooney | 5,856 Pediatric Images

5,856	4,173	1,043	624	73%
Total Images	**Training Set**	**Validation Set**	**Test Set**	**Pneumonia**
All splits	80% stratified	20% stratified	Original held-out	Train imbalance



Methodology Overview

Experimental & quantitative design — identical conditions for all models

Dataset Prep
5,856 images
Stratified 80/20 split

Preprocessing
RGB → 224×224
Model-specific normalise

Augmentation
Rotation ±15°, Shift, Shear
Zoom, H-Flip (train only)

Model Training
CNN + EfficientNetB0
+ ResNet50 (2-phase)

Evaluation
Acc, Prec, Recall
F1, AUC-ROC

Grad-CAM & Error Analysis
Explainability
+ FP/FN patterns

Training Strategy:

All Models

- Binary Cross-Entropy loss
- Adam optimiser
- Class weighting (balanced)
- EarlyStopping (patience=7)
- ReduceLROnPlateau (patience=3)
- ModelCheckpoint (best val_AUC)
- Random seed=42 (reproducible)

Custom CNN

- 4 convolutional blocks
- Filters: 32 → 64 → 128 → 256
- Batch Norm + Dropout (0.25–0.50)
- L2 regularisation (1e-4)
- Global Average Pooling
- Dense(256) → Sigmoid output
- Max 50 epochs, LR=1e-3

Transfer Learning (EfficientNetB0 & ResNet50)

- Phase 1 — Feature Extraction: backbone frozen, LR=1e-3, 10 epochs
- Phase 2 — Fine-Tuning: last 10 layers unfrozen, LR=1e-6, 12 epochs
- Classification head: GAP → BN → Dense(256) → Dropout(0.55) → Sigmoid

Implementation Workflow

Kaggle Notebook walkthrough — Python / TensorFlow / Keras / Scikit-learn

1 Setup & Imports

TensorFlow, Keras, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, OpenCV, PIL

2 Data Exploration

Count images per split / Plot class distribution / Visualize sample X-rays (Figures 2-3)

3 Preprocessing & Data Augmentation

RGB -> 224x224 -> Normalize [0,1] / ImageDataGenerator: Rot+-15deg, Shift, Shear, Zoom, H-Flip / Stratified 80/20 split (seed=42)

4 Custom CNN Training

build_custom_cnn() -> 4 Conv blocks (32/64/128/256 filters) / EarlyStopping + ReduceLROnPlateau + ModelCheckpoint / class_weight passed to fit()

5 Transfer Learning Training

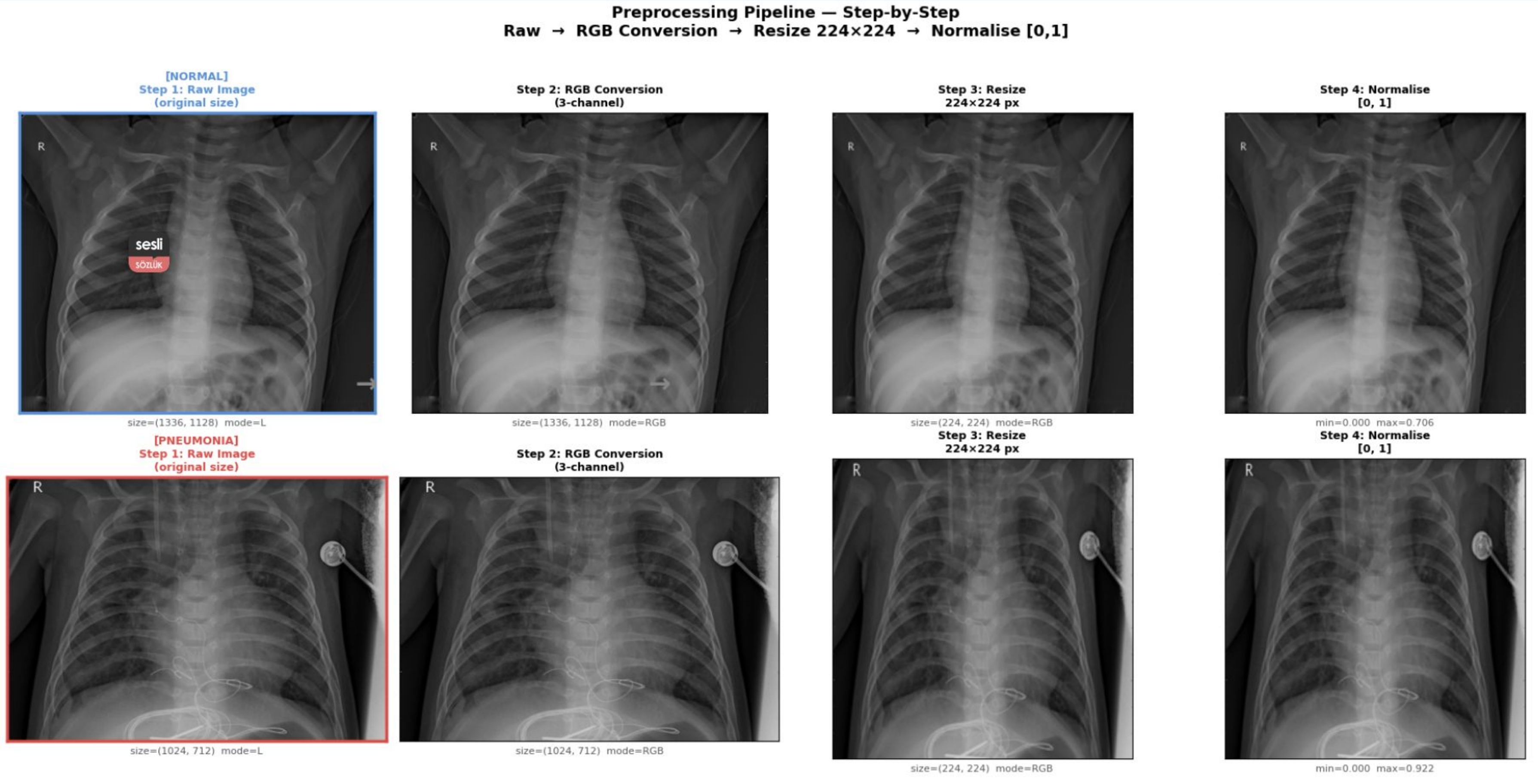
Phase 1: backbone frozen, LR=1e-3, 10 epochs / Phase 2: last 10 layers unfrozen, LR=1e-6, 12 epochs / Same strategy for EfficientNetB0 and ResNet50 as well

6 Evaluation & Explainability

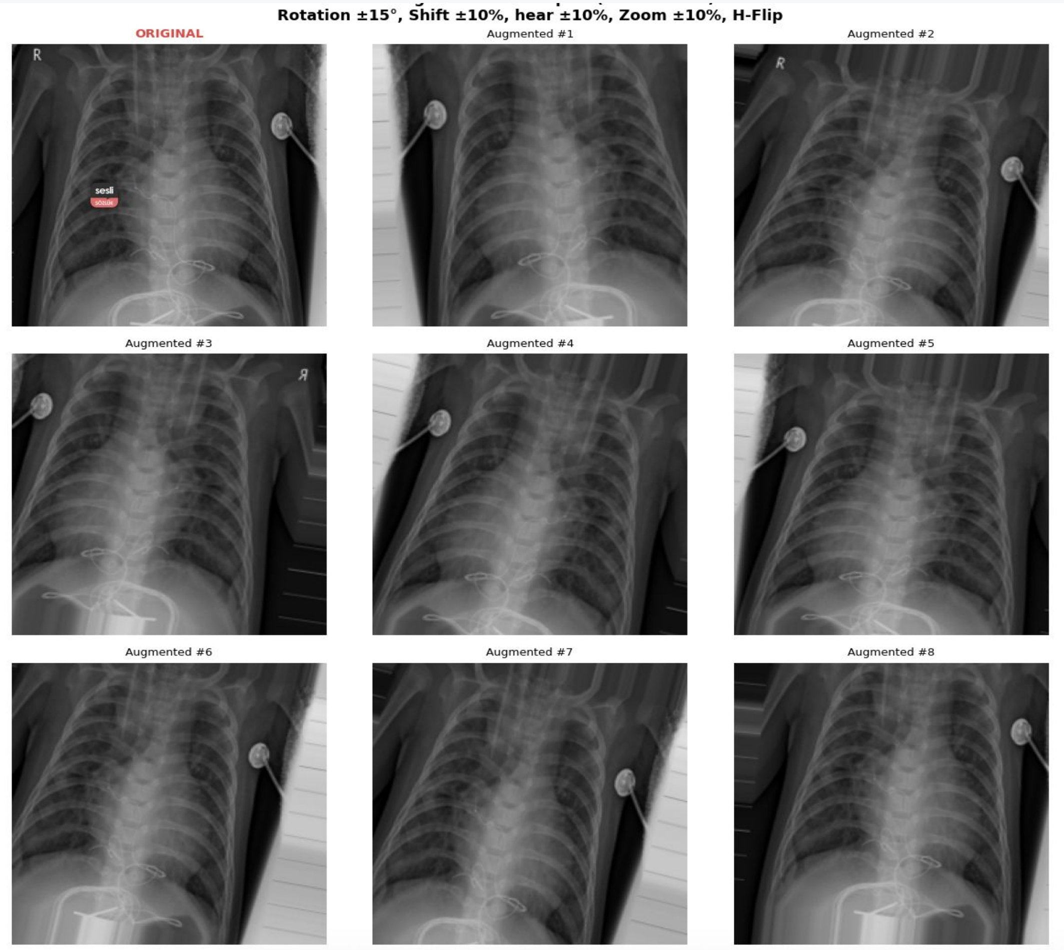
Confusion matrix / ROC-AUC curves / GradCAM.compute_heatmap() + overlay() / Error analysis: FP and FN image grids + confidence distributions

All outputs are saved to /outputs/figures/, /tables/, /models/ | Full notebook on Kaggle + GitHub

Preprocessing Pipeline

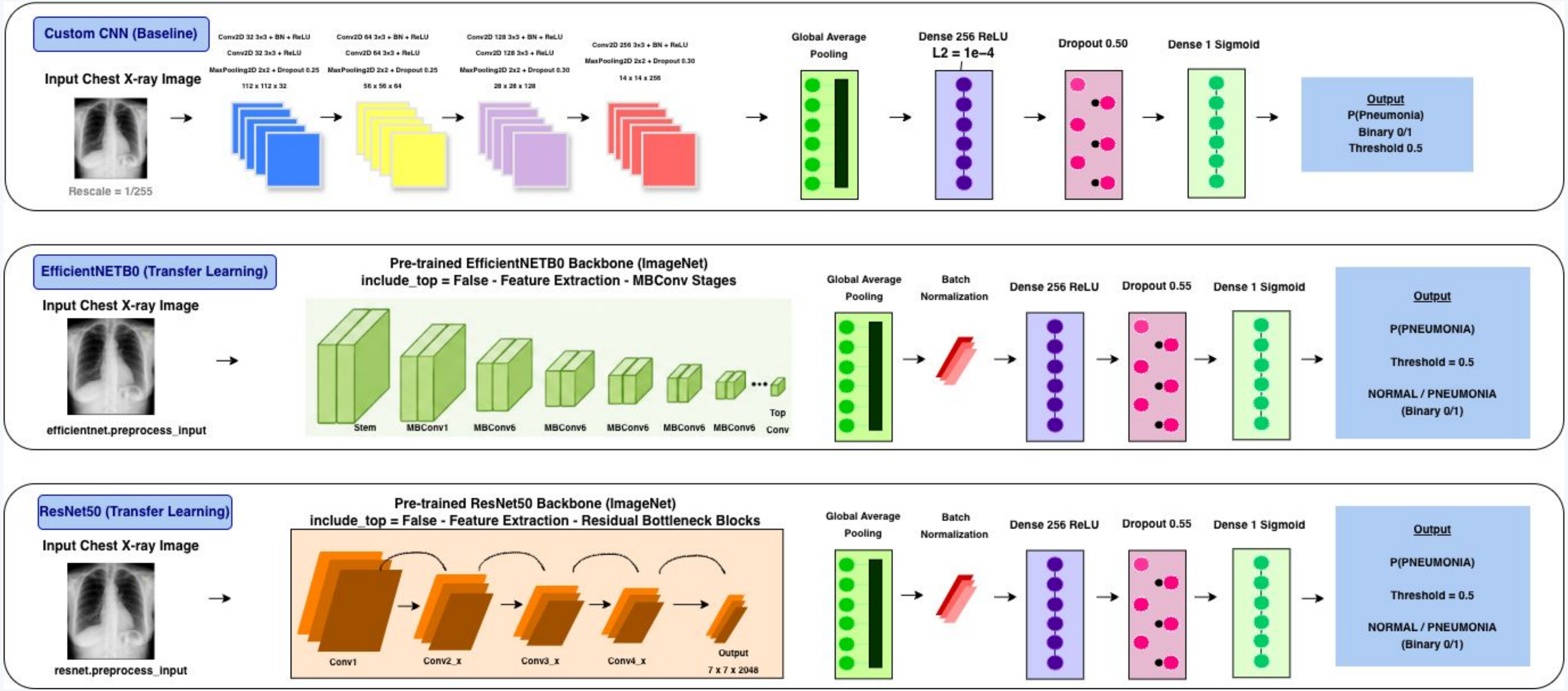


Augmentation Samples



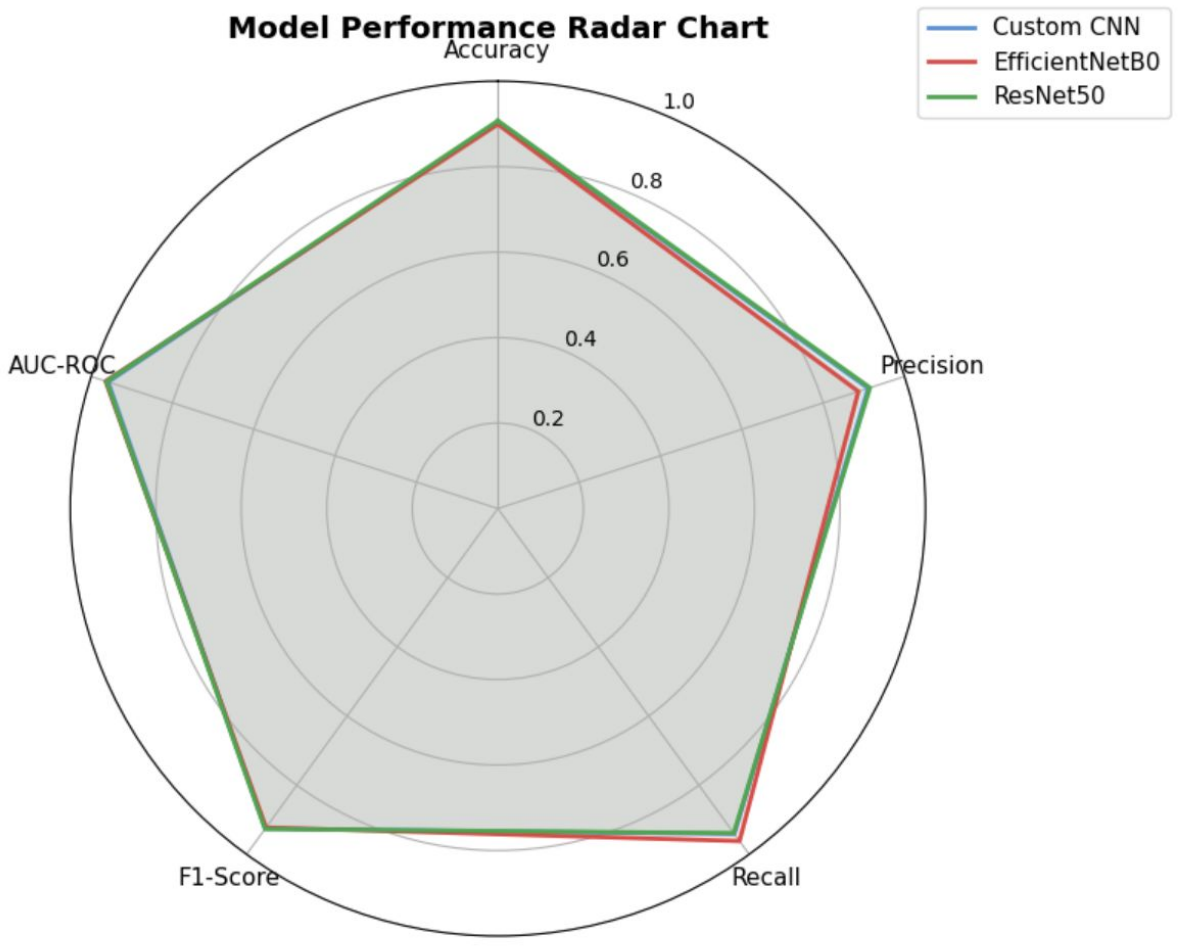
Model Architectures

Three architectures compared under identical experimental conditions



Model Comparison Chart

Radar Chart



Model Performance Radar Chart
Custom CNN
EfficientNetB0
ResNet50
Accuracy
Precision
Recall
F1-Score
AUC-ROC
0.2
0.4
0.6
0.8
1.0

Results: Overall Model Performance

Test set — 624 images (234 Normal + 390 Pneumonia)

Model	Accuracy	Precision	Recall ↑	F1-Score	AUC-ROC
Custom CNN	0.9054	0.9107	0.9410	0.9256	0.9583
EfficientNetB0	0.8990	0.8865	**0.9615**	0.9225	**0.9645**
ResNet50	**0.9071**	**0.9150**	0.9385	**0.9266**	0.9623

Key Clinical Findings

EfficientNetB0

Recall: 96.2%

Only 15 missed pneumonia cases out of 390 — best for false negative reduction

ResNet50

F1: 0.9266, Acc: 90.7%

Most balanced profile across all metrics — best overall accuracy and F1

Custom CNN

AUC: 0.9583

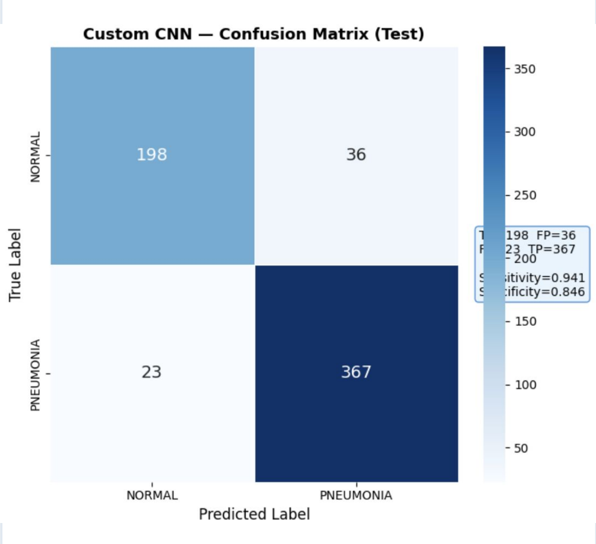
Competitive with transfer models despite training from scratch; good explainability

⚠️ In pneumonia screening, Recall is the most critical metric because in case of missed cases (False Negatives) delays treatments and directly harms patients.

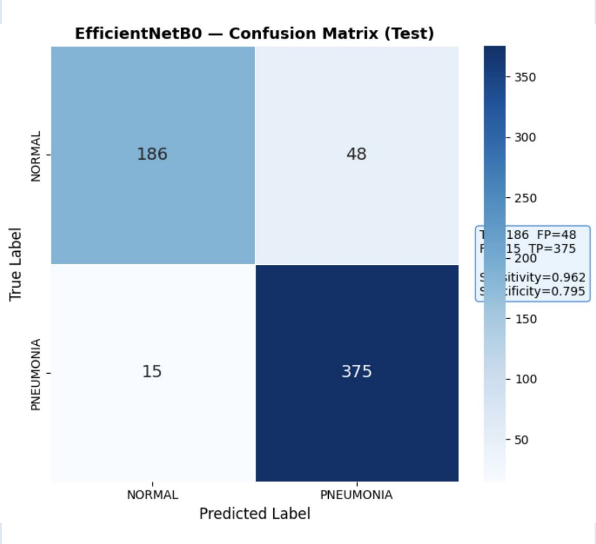
Class-Level Analysis: Confusion Matrices & ROC Curves

False Positive vs False Negative trade-offs for each model

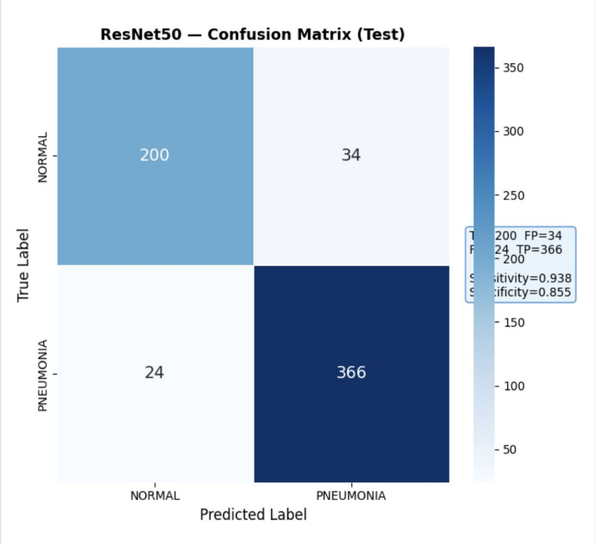
Custom CNN



EfficientNetB0



ResNet50



ROC / AUC Summary

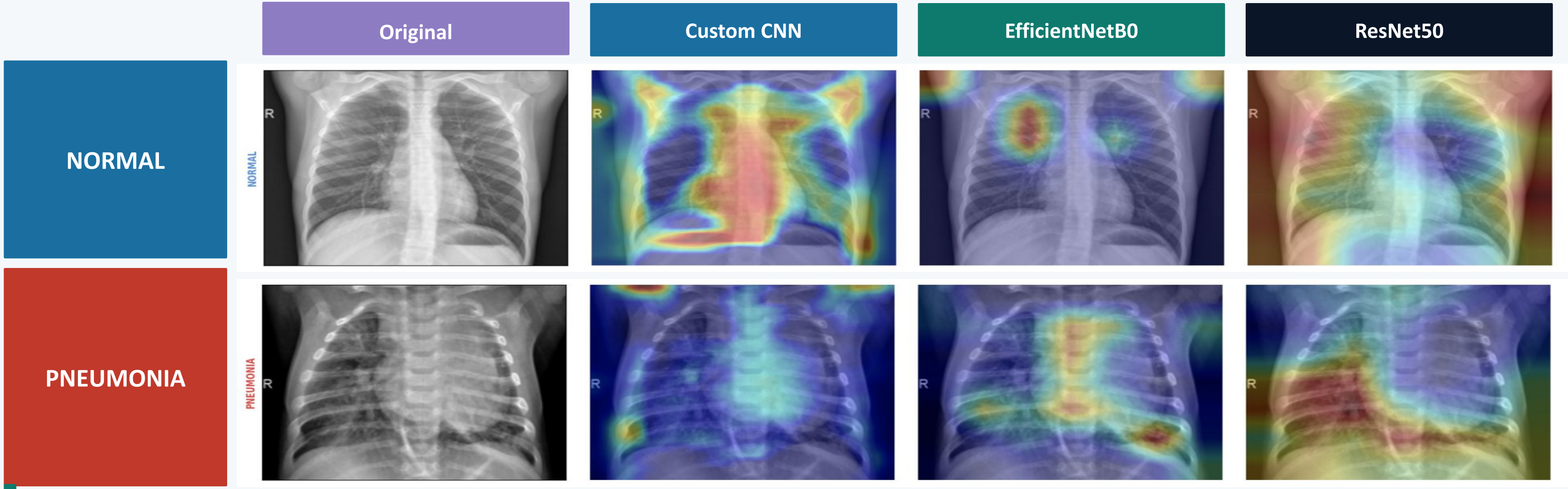
- Custom CNN AUC = 0.9583
- EfficientNetB0 AUC = 0.9645 ★
- ResNet50 AUC = 0.9623

Grad-CAM Explainability (RQ5)

Gradient-weighted Class Activation Mapping — which image regions drive each prediction?

Grad-CAM uses gradients from the last convolutional layer to produce a heatmap highlighting the spatial regions most influential in the model's prediction and given as overlaid on the original image.

Grad-CAM Comparison Grid (Figures 16–22 from Report)



Custom CNN ✓ Clinically coherent activations which are focuses on lung parenchyma and consolidation regions; most interpretable

Transfer Models ⚠️ Sigmoid saturation (output → 1.0000) causes gradients to vanish before reaching the target layer ($\sigma'(x) \to 0$). Mitigation attempted: logit bypass + GuidedGradCAM on last backbone conv layer. Result: heatmaps improved marginally but remained diffuse — attributed to limited fine-tuning scope (last 10 layers only)

Error Analysis (RQ7)

Misclassification patterns across all three models

Custom CNN

Total Errors:	59
False Positives (Normal → Pneumonia):	FP = 36
False Negatives (Pneumonia → Normal):	FN = 23

EfficientNetB0

Total Errors:	63
False Positives (Normal → Pneumonia):	FP = 48
False Negatives (Pneumonia → Normal):	FN = 15

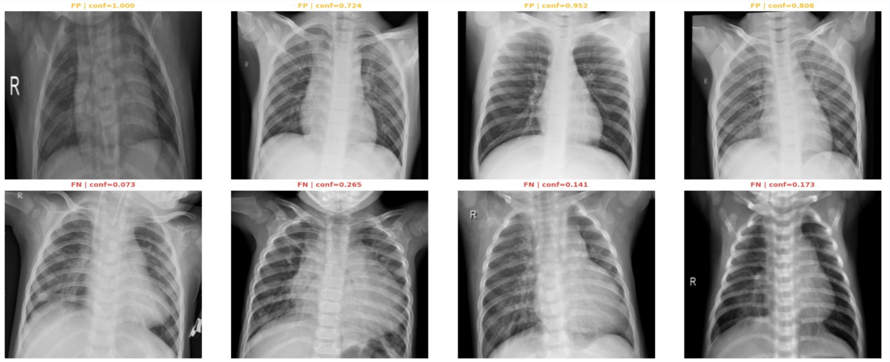
ResNet50

Total Errors:	58
False Positives (Normal → Pneumonia):	FP = 34
False Negatives (Pneumonia → Normal):	FN = 24

Representative Error Cases for Custom CNN

False Positives — Normal images misclassified as Pneumonia

False Negatives — Pneumonia images missed as Normal



- False Positives: Normal images with artifact-like opacities or challenging lung anatomy — misclassified with high confidence (0.7–1.0)
- False Negatives: Subtle, low-contrast Pneumonia findings where models show uncertainty (0.0–0.4); threshold reduction could improve recall
- 🚩 High-confidence wrong predictions (e.g. FP conf=1.000 in Custom CNN) signal possible shortcut learning → threshold calibration needed before clinical deployment

Research Question Answers

All seven RQs answered through quantitative evidence

RQ1	**Custom CNN effectiveness?**	90.5% accuracy, 94.1% Pneumonia recall, AUC=0.958	✓
RQ2	**Transfer learning vs custom CNN?**	EfficientNetB0 leads on Recall (96.2%) & AUC (0.965); ResNet50 leads Accuracy & F1; all within 1% of each other	✓
RQ3	**Class weighting on false negatives?**	EfficientNetB0 achieved lowest FN=15 vs 23–24 for others. Class weights: Normal=1.94, Pneumonia=0.67	✓
RQ4	**Multi-metric evaluation impact?**	Accuracy alone masked class-level differences. Recall and AUC most discriminating for pneumonia detection	✓
RQ5	**Grad-CAM model insights?**	Custom CNN: clinically coherent lung-region activations. Transfer models: diffuse activations due to sigmoid saturation	✓
RQ6	**Data augmentation on generalisation?**	Stable convergence, no severe overfitting across all three models. Fine-tuning phase improved validation AUC	✓
RQ7	**Misclassification patterns?**	FP: artifact-like opacities with high confidence. FN: subtle low-contrast findings with low-mid confidence	✓

Limitations & Future Research Directions

Honest assessment and a clear path forward

Current Limitations

- Single public dataset — no external validation across hospitals or demographics
- Grad-CAM sigmoid saturation in transfer models — GuidedGradCAM did not fully resolve diffuse activations
- Limited hyperparameter optimisation — only last 10 backbone layers unfrozen
- No SHAP or complementary gradient-free explainability methods
- No real-world clinical deployment testing or prospective validation
- Binary labels only — bacterial vs viral pneumonia subtype distinction not possible

Future Research Directions

- **External validation:** Multi-centre datasets for generalisability across patient populations
- **Deeper fine-tuning:** Full backbone unfreeze + ScoreCAM / SHAP for explainability
- **Multi-class extension:** Bacterial vs viral pneumonia subtype classification
- **Alternative architectures:** DenseNet121, MobileNetV2 for mobile/edge deployment
- **Semi-supervised learning:** Reduce labelling cost with unlabelled data
- **Prospective clinical study:** Randomised controlled trial with real radiologist integration

Ethical Considerations

AI systems must act as radiologist support tools — not autonomous decision-makers. Data privacy preserved (anonymised Kaggle dataset). Any clinical deployment requires prospective validation and regulatory approval.

Conclusion

1

Competitive performance across all models

All three architectures exceeded 89% accuracy and 0.958 AUC — demonstrating the viability of deep learning for automated pneumonia screening.

2

EfficientNetB0 — best for clinical recall

96.2% Pneumonia recall (only 15 missed cases) makes it the strongest candidate where minimising false negatives is paramount.

3

Class weighting is effective

Balanced class weights measurably reduced false negative rates — essential for handling the 73%/27% imbalance without data resampling.

4

Explainability gap is a real concern

Custom CNN produced clinically coherent Grad-CAM maps; transfer model sigmoid saturation limited heatmap quality — deeper fine-tuning and gradient-free methods needed.

5

Reproducible, integrated pipeline

Systematic multi-model comparison + class weighting + Grad-CAM + error analysis in a single Kaggle Notebook — publicly available on GitHub.

6

Clinical Deployment Guidance

For settings where minimising missed cases is paramount ⟶ EfficientNetB0 (Recall: 96.2%). For settings requiring balanced performance across all metrics ⟶ ResNet50 (F1: 0.927, Acc: 90.7%). Both require threshold calibration and prospective validation before clinical use

Thank you.